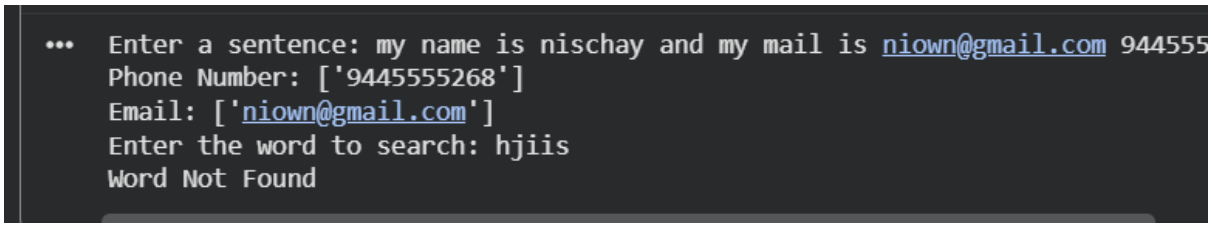
EXP.1. Write program demonstrates how to use regular expressions in Python to match and search for patterns in text.

CODE

```
import re

text = input("Enter a sentence: ")

phone = re.findall(r"\d{10}", text)

email = re.findall(r"\S+@\S+", text)

print("Phone Number:", phone)

print("Email:", email)

word = input("Enter the word to search: ")

if re.search(word, text):

    print("Word Found")

else:

    print("Word Not Found")
```

Output:

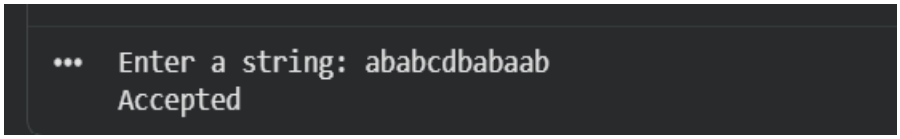


EXP.2. Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we'll create a simple automaton to match strings ending with 'ab' using python.

CODE:

```
def finite_state_automaton(string):
    return string.endswith("ab")
s = input("Enter a string: ")
if finite_state_automaton(s):
    print("Accepted")
else:
    print("Rejected")
```

OUTPUT:

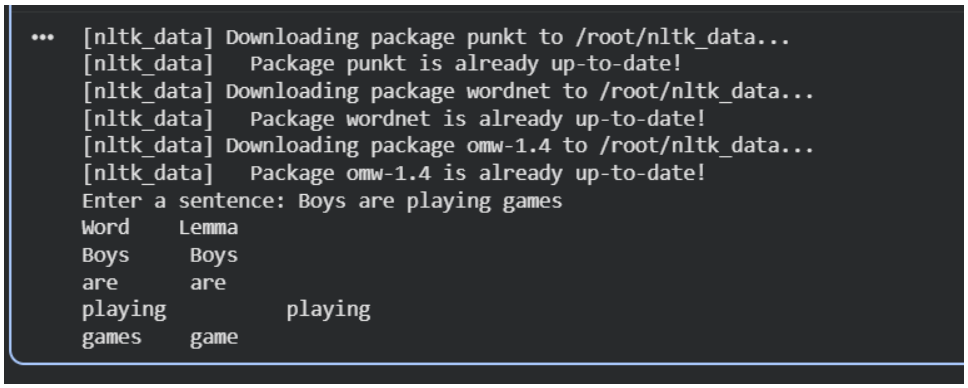


EXP.3. Write program demonstrates how to perform morphological analysis using the NLTK library in Python.

CODE:

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('omw-1.4')
text = input("Enter a sentence: ")
words = word_tokenize(text)
lemmatizer = WordNetLemmatizer()
print("Word\tLemma")
for word in words:
    print(word, "\t", lemmatizer.lemmatize(word))
```

OUTPUT:

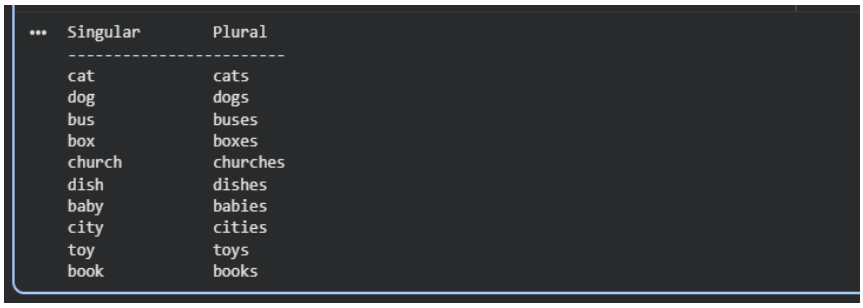


EXP.4 Implement a finite-state machine for morphological parsing. In this example, we'll create a simple machine to generate plural forms of English nouns using python.

CODE:

```
def generate_plural(noun):
    if noun.endswith(("s", "x", "z", "ch", "sh")):
        plural = noun + "es"
    elif noun.endswith("y") and len(noun) > 1 and noun[-2].lower() not in "aeiou":
        plural = noun[:-1] + "ies"
    else:
        plural = noun + "s"
    return plural
words = ["cat", "dog", "bus", "box", "church",
         "dish", "baby", "city", "toy", "book"]
print("Singular\tPlural")
print("------------------------")
for word in words:
    print(f"{word:10}\t{generate_plural(word)}")
```

OUTPUT:

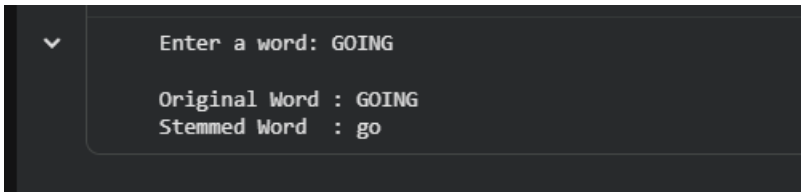


EXP.5. Use the Porter Stemmer algorithm to perform word stemming on a list of words using python libraries.

CODE

```
import nltk
from nltk.stem import PorterStemmer
stemmer = PorterStemmer()
word = input("Enter a word: ")
stem = stemmer.stem(word)
# Display result
print("\nOriginal Word :", word)
print("Stemmed Word  :", stem)
```

OUTPUT:



EXP.6. Implement a basic N-gram model for text generation. For example, generate text using a bigram model using python.

CODE:

```
import nltk
from nltk import bigrams
from collections import defaultdict
import random
nltk.download('punkt')
nltk.download('punkt_tab')
text = input("Enter a sentence: ")
words = nltk.word_tokenize(text.lower())
model = defaultdict(list)
for w1, w2 in bigrams(words):
    model[w1].append(w2)
start = input("Enter the starting word: ").lower()
if start not in model:
    print("Starting word not found in the sentence.")
else:
    generated = [start]
    current = start
    for i in range(10):
        if current not in model:
            break
        current = random.choice(model[current])
        generated.append(current)
    print("\nGenerated Text:")
    print(" ".join(generated))
```

OUTPUT:

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
Enter a sentence: MY NAME IS NISCHAY
Enter the starting word: MY

Generated Text:
my name is nischay
```